

The Signaling Gap

State-Owned Media and the Strategic Management of
Sentiment during the Russian Invasion of Ukraine

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The Puzzle

Constrained Communication

When exit is too costly and direct voice risks retaliation, actors seek indirect channels to communicate dissatisfaction.

- **Hirschman (1970):** Exit, Voice, and Loyalty — when exit and voice are both foreclosed, actors search for alternative channels
- **Kassing (1997):** Organizational dissent — subordinates route disagreement to external audiences who have leverage over the principal
- Logic is general: any hierarchical relationship where a subordinate depends on a patron but must also manage other audiences

In international politics, the stakes are highest.

When Formal Channels Become Liabilities

States that depend on a powerful patron for their security **do not always have the luxury of saying what they think.**

- Formal channels (UN votes, official statements) risk retaliation from the patron
- But **silence risks being seen as endorsement** when the patron draws international condemnation

Question: What channels do constrained states use to signal distance from a patron's position, and when are they credible?

A Gap Between Scholars and Practitioners

Political Science

State media studied as a **domestic** tool:

- Agenda setting (McCombs 2002)
- Spin dictators (Guriev & Treisman 2022)
- Autocratic media management (Egorov et al. 2009)

Orientation: how governments shape what *their own publics* think

Intelligence Practice

State media monitored as **international signal**:

- BBC Monitoring (est. 1939)
- FBIS / Kremlinology (Cold War)
- OSINT formalized (Manor 2019; USIC 2024)

Orientation: reading state media as a window into *regime preferences*

Same outlet does both. Tools exist — but haven't been assembled.

Argument: State-owned media provides an institutionally mediated signaling channel through which constrained states signal distance from their patron's position to external audiences.

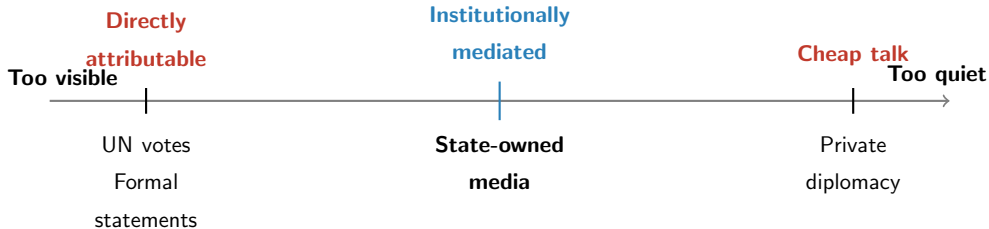
Preview of findings:

1. The invasion produced a significant negative shift in media sentiment toward Russia across the region
2. State-owned media shifted *less* negatively than independent media → strategic management
3. State media in Russia-*aligned* states shifted *more* negatively than neutral states — **opposite of domestic suppression prediction**

These states refused to formally condemn the invasion while allowing their state media to turn sharply critical.

Theory

Why State Media Rather Than Alternatives?



- Public enough to be observed by foreign governments that systematically monitor it
- Below the formal-break threshold that triggers patron retaliation
- Readable as a regime choice because external observers know state media reflects regime preferences

Hypotheses

- H1:** An internationally condemned action by the patron produces a significant decline in media sentiment toward the patron across both state-owned and independent outlets.
- H2:** State-owned media will report negative sentiment toward the patron after the shock to a lesser extent than independent media.
- H3:** State-owned media in states formally allied to the patron will respond more negatively to the shock than state-owned media in neutral states.

Hypotheses

H1: An internationally condemned action by the patron produces a significant decline in media sentiment toward the patron across both state-owned and independent outlets.

Without a baseline negative shock, the divergence in H2 and H3 would be uninterpretable.

H2: State-owned media will report negative sentiment toward the patron after the shock to a lesser extent than independent media.

H3: State-owned media in states formally allied to the patron will respond more negatively to the shock than state-owned media in neutral states.

Hypotheses

H1: An internationally condemned action by the patron produces a significant decline in media sentiment toward the patron across both state-owned and independent outlets.

H2: State-owned media will report negative sentiment toward the patron after the shock to a lesser extent than independent media.

A differential response establishes that state media operates under political constraints — this is what makes coverage readable as a regime choice for H3.

H3: State-owned media in states formally allied to the patron will respond more negatively to the shock than state-owned media in neutral states.

Hypotheses

H1: An internationally condemned action by the patron produces a significant decline in media sentiment toward the patron across both state-owned and independent outlets.

H2: State-owned media will report negative sentiment toward the patron after the shock to a lesser extent than independent media.

H3: State-owned media in states formally allied to the patron will respond more negatively to the shock than state-owned media in neutral states.

Domestic suppression and signaling generate opposite predictions — this is the critical test.

Competing Predictions for H3

Domestic Suppression

- Patron-aligned states **suppress** negative coverage to protect alliance
- State media *less* negative than neutral states
- Alternative: Higher ethnic Russian pop. → **more** negative

Signaling

- Patron-aligned states use state media to **signal distance** because formal channels are foreclosed
- State media *more* negative than neutral states
- Ethnic Russian pop. → **no effect** (logic is security alignment, not demographics)

Opposite predictions provide leverage for distinguishing between mechanisms.

Data & Design

Sample and Measurement

Coverage: Aug 2021 – Aug 2022

Articles: ~174,000 (excl. Ukraine)

Observation: Publication Date-Time

Sources: 61 outlets, 11 countries

Languages: Neural machine translation

Russia-Aligned	Armenia, Azerbaijan, Belarus, Kazakhstan
Neutral	Georgia, Moldova, Kosovo, Serbia
NATO	Albania, Hungary, Turkey

GPT-4o classification (title + first two sentences):

1. **Topic:** Political, Economic, Military, Cultural, Other
2. **Russia involvement:** Binary
3. **Sentiment toward Russia:** Ternary (−1, 0, +1)

Validation: 100 adversarially double-coded articles

F1: 0.92 sentiment, 0.80 topic

Identification Strategy

Core approach: Regression Difference-in-Discontinuity in Time

- February 24 as a sharp break in the information environment
- Troop buildup was gradual; full-scale invasion was discrete and unanticipated in scale
- Running variable: “Probabilistic” Publication Time (measured at date-time)

Within-group RDDs are causal; cross-group comparisons are correlational but subject to extensive robustness checks. Where missing, publication time is replaced by a source or country level KDE. Findings robust to alternative time specifications.

▸ Specifications

▸ Z-Score

▸ Inference

▸ DAG

▸ Pre-Trends

Identification Strategy

Three levels of analysis:

1. **H1 — RDD:** Effect of invasion on sentiment (all media)
2. **H2 — Diff-in-Disc:** State-owned $\times$ Treatment interaction
3. **H3 — Diff-in-Disc:** Russia-aligned $\times$ Treatment interaction (state media only)

Bandwidth selection via `rdrobust`. Source-level Z-score normalization or source/state/topic fixed effects. Clustering at the source level; wild cluster bootstrap for few-cluster settings.

► Specifications

► Z-Score

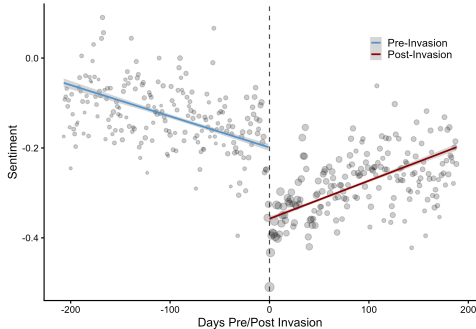
► Inference

► DAG

► Pre-Trends

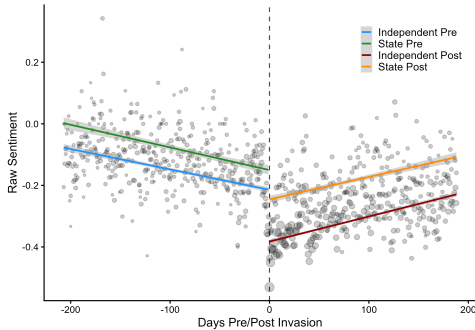
Results

H1: The Invasion Produced a Negative Sentiment Shock



► Full Specifications

H2: State Media Managed Coverage More Carefully



Both shift negative, but independent media shifts significantly more. Consistent with strategic management by state outlets.

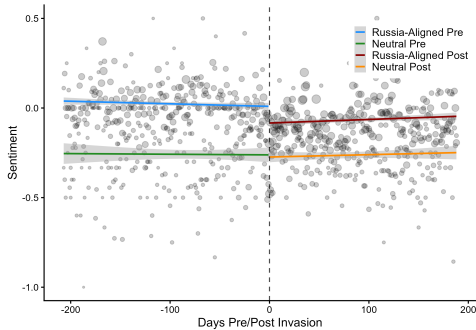
► Full Specifications

► By Topic

► Floor Effect

► Serbia Pair

H3: Russia-Aligned State Media Shifted More Negatively



Russia-aligned state media (blue/red) starts higher and drops more sharply than neutral state media (green/orange). Opposite of domestic suppression.

► Full Specifications

Putting It Together

Hypothesis	Prediction	Result
H1: Baseline shock	Negative shift	Confirmed (−0.11 to −0.28 SD)
H2: State vs. Independent	State less negative	Confirmed (+0.22 SD)
H3: Russia-aligned vs. Neutral	Aligned <i>more</i> negative	Confirmed (−0.33 to −0.38 SD)

- Pattern is *opposite* of what domestic audience management predicts
- Alternative partition by ethnic Russian population produces no differential — rules out the demographic mechanism ▶ Ethnic Partition
- Robust to bandwidth, matching, topic restrictions, leave-one-out, placebo

Contributions

Contributions

1. Bridging the theory-practice gap

- Signals practitioners monitored for 80+ years are theoretically structured and empirically identifiable
- Not idiosyncratic — follows from the logic of alliance constraints

2. Institutionally mediated signaling under alliance constraints

- When formal channels are foreclosed, states turn to institutionally mediated ones with predictable patterns
- General framework: any patron-client relationship under constraint

3. Methodological

- GPT-4o classification at scale (174K articles, $F1 = 0.92$)
- RD in time for media analysis — template for future work
- Matched sensitivity procedure for difference-in-discontinuities — coarsened exact matching paired with Cinelli & Hazlett (2020) [▶ Sensitivity](#)

Thank You

Appendix

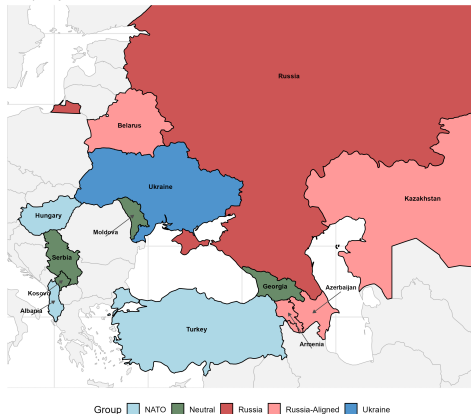
A. Theory and Case

February 24, 2022

The Russian invasion of Ukraine forced Russia's security partners to navigate this bind.

- Russia-aligned states had reasons to oppose the invasion
- Violated sovereignty norms that protect small states
- Threatened substantial Western trade ties
- But formal condemnation of a patron willing to use force → not viable

Study Region: Russia's Near Abroad



Signaling Under Constraint

Standard signaling theory: credible signals require costs (Fearon 1997)

- Audience costs: leaders stake political capital on public positions
- Sunk costs: troop deployments, sanctions, arms transfers
- Common feature: **visible, attributable, formal** channels

But alliances constrain:

- Alliance membership trades autonomy for security (Morrow 1991)
- Public dissent risks abandonment (Snyder 1997)
- Formal channels lock precisely when dissent is most needed (Weitsman 2004)

→ **What do constrained states do when formal channels are unavailable?**

The Signaling Mechanism

State media signals work through two properties:

1. Readable

- Common knowledge of state control → deviation from baseline is informative (Gehlbach et al. 2014)
- Foreign analysts read shifts as regime choice, not journalistic noise
- Why not just loosen independent media? Because independent shifts aren't readable as a *regime choice* — only state media carries the attribution

The Signaling Mechanism

State media signals work through two properties:

2. Costly → Belief-Shifting

- Allowing negative coverage of a security patron risks provoking retaliation
- A state with no reason to signal distance would not accept this risk → bluffing is irrational
- The signal shifts the receiver's beliefs about alignment — it doesn't require revealing "true" preferences, only that foreign analysts assign higher probability to non-alignment after observing the shift

Costs, Benefits, and Sustainability

The strategic calculus:

- **Cost of signaling:** Risk of patron retaliation — Russia punished Georgia (2008) and Ukraine (2014) for formal breaks
- **Benefit of signaling:** Shifting Western beliefs about alignment → sanctions treatment, diplomacy, trade preservation
- **Cost of not signaling:** Being categorized alongside an internationally condemned patron

Costs, Benefits, and Sustainability

Sustainability (scope condition — even a one-shot signal is valuable):

- **Formal-break threshold:** State media stays below the line that triggers patron punishment. Institutional separation between government and editorial function provides deniability — coverage can be disavowed as editorial judgment rather than state policy (Entman 2004)
- **Mutual dependence:** Russia values CSTO for strategic reasons (Allison 2008). Punishing a client state for media-level dissent risks pushing that state toward the West — precisely the outcome Russia's alliance framework is designed to prevent

B. Data and Measurement

Measurement: GPT-4o Classification

Three dimensions coded at article level (title + first two sentences):

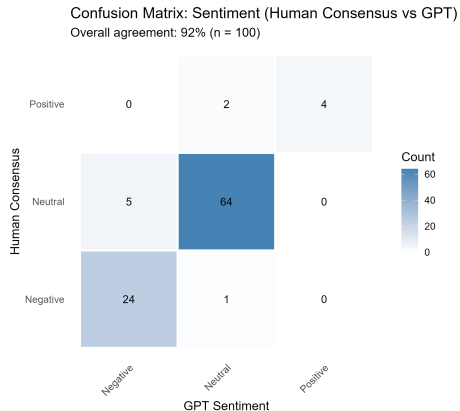
1. **Topic:** Political, Economic, Military, Cultural, Other
2. **Russia involvement:** Binary (main actor or not)
3. **Sentiment toward Russia:** Ternary (-1 , 0 , $+1$)

Validation: 100 adversarially double-coded articles (coauthors debated disagreements)

Dimension	GPT F1 Score
Sentiment	0.92
Topic	0.80

Classifier biases (over-classifying “Other” for topic; under-classifying Russia involvement) both bias toward null. [► Confusion matrix](#)

GPT Classifier: Sentiment Confusion Matrix



92% overall agreement ($n = 100$). Adversarial double-coding by coauthors with consensus resolution. Misclassification biases toward null.

What Do These Articles Look Like?

Positive Sentiment Toward Russia

Country: Belarus

Outlet: *eng.belta.by* (State-owned)

Date: 2022-04-12

"Lukashenko: Belarus has something to contribute to Russia's might."

"Belarus and Russia will surely endure in the current circumstances. Belarus President Aleksandr Lukashenko expressed confidence about it during negotiations with President of Russia Vladimir Putin at the Vostochny Cosmodrome in Russia's Amur Oblast on 12 April."

What Do These Articles Look Like?

Neutral Sentiment Toward Russia

Country: Armenia

Outlet: *golosarmenii.am* (Independent)

Date: 2022-04-01

“Armenia abstained in the UNHRC vote on unilateral sanctions.”

“Twenty-seven of the 47 countries, including Russia, China and India, voted in favour of the relevant resolution adopted at the meeting on Thursday, while 14 States, including the United States, the European Union, the United Kingdom and Ukraine, objected. Six delegations abstained in the vote, including Armenia.”

What Do These Articles Look Like?

Negative Sentiment Toward Russia

Country: Serbia

Outlet: *danas.rs* (Independent)

Date: 2022-05-26

“De-Ukrainization in Mariupol: Russians installed mobile TV propaganda, children abolished holidays.”

“The Russian forces that occupied Mariupol have decided to cancel the summer vacation so that the students can master the Russian program, said today the mayor’s adviser Petro Andryushchenko. Children will have classes in Russian language and literature, Russian history and mathematics all summer.”

Same War, Same Country, Opposite Orientation

Independent — N1, February 25 coded **negative**

“Canak: Putin started the attack on the same day that Hitler presented his program”

A Serbian politician's charge that Putin launched the invasion on the anniversary of Hitler's Nazi program.

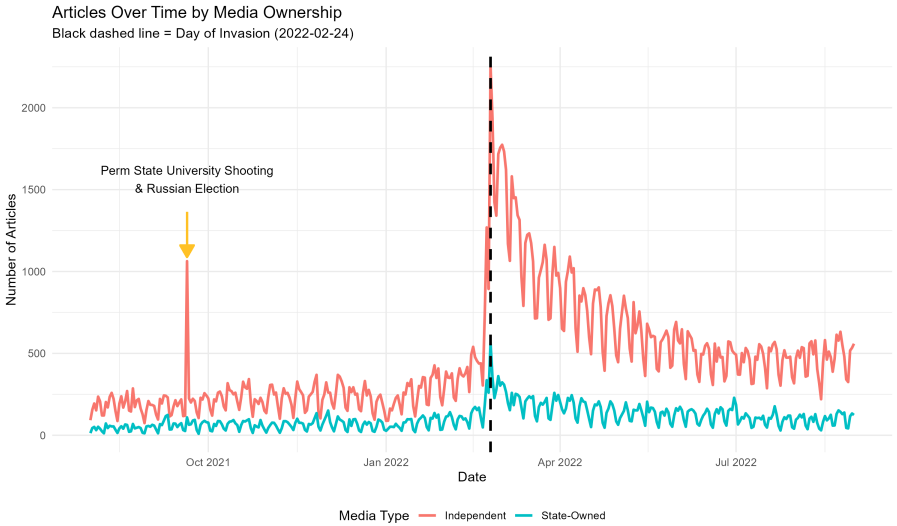
State-owned — RTS, February 26 coded **neutral**

“Bocan-Harchenko: Russia understands Serbia's position, does not require anything from Belgrade”

A report built around the Russian ambassador's assurances to Belgrade.

The medium is held constant — both Serbian outlets covering the invasion in real time — so the gap in orientation tracks ownership rather than the event. Translations are machine-generated.

Article Volume Over Time



Ethnic Russian Population × Security Alignment

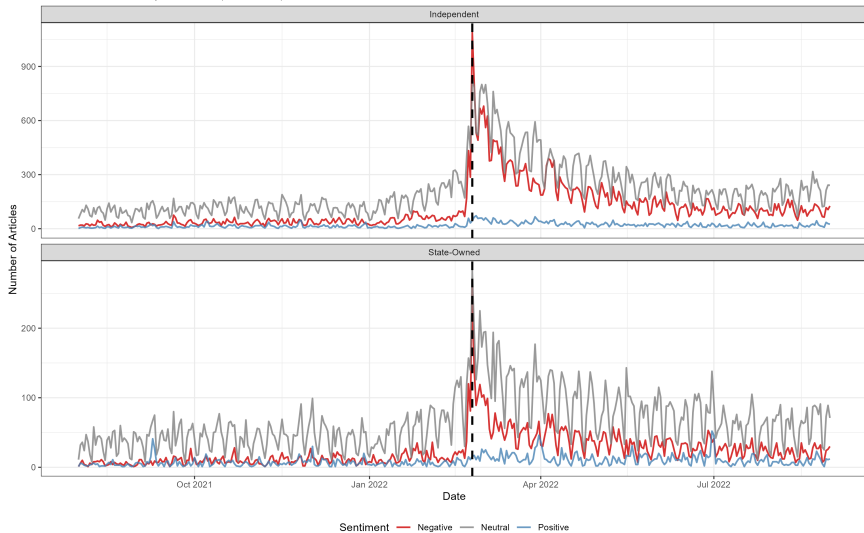
	Neutral	Russia-Aligned
Low	Kosovo, Serbia	Azerbaijan, Armenia
High	Moldova, Georgia	Belarus, Kazakhstan

Orthogonal variation enables separate tests of signaling vs. domestic audience mechanisms. [► Results](#)

Daily Sentiment Counts by Media Type

Daily Sentiment Article Counts by Media Type

Black dashed line = Day of Invasion (2022-02-24)



C. Research Design

Specification: Base RDD (H1)

$$Y_p^z = \beta_0 + \beta_1 D_p + \tau T_p + \beta_3 D_p T_p \\ + \beta_4 G_p + \beta_5 T_p G_p + \beta_6 D_p G_p + \beta_7 D_p T_p G_p + \gamma_{\text{topic}} + \varepsilon_p$$

Y_p^z Z-Score Normalized Sentiment

D_p Running variable (probabilistic date-time)

T_p Post-invasion indicator

τ **Coefficient of interest:** effect of the invasion

G_p Group indicator (introduced in H2/H3)

Bandwidth selected via `rdrobust`. SEs clustered at source level.

Specification: Diff-in-Disc (H2, H3)

$$Y_p^z = \beta_0 + \beta_1 D_p + \tau T_p + \beta_3 D_p T_p \\ + \beta_4 G_p + \beta_5 \mathbf{T}_p \mathbf{G}_p + \beta_6 D_p G_p + \beta_7 D_p T_p G_p + \gamma_{\text{topic}} + \varepsilon_p$$

G_p Group indicator

β_5 **Coefficient of interest:** differential treatment effect

H2: $G_p = \text{State-owned (full sample)}$

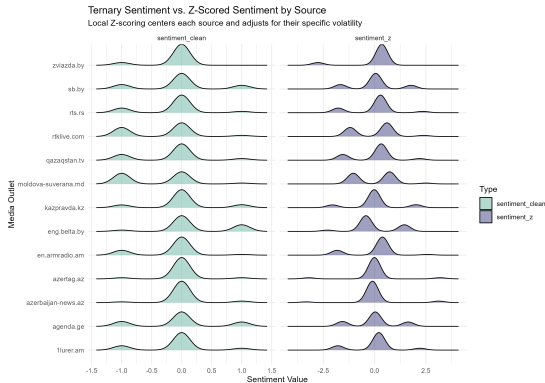
H3: $G_p = \text{Russia-aligned (state media only)}$

WCR bootstrap for interaction inference ($B = 99,999$, Webb weights). Z-score normalization.
Optimal bandwidth via `rdrobust`.

Source-Level Z-Score Normalization

Problem: Outlets have different baseline sentiment levels and volatility.

Solution: Z-score normalizes each source to its own pre-invasion distribution.



Inference with Few Clusters

Challenge: Source-level clustering yields ~ 13 –61 clusters.

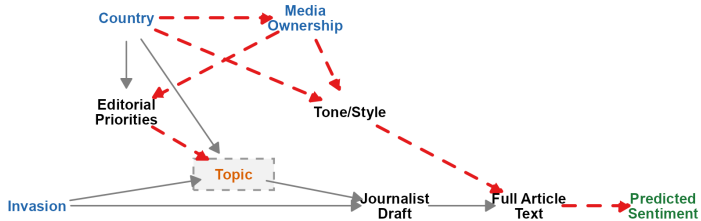
Approach:

- H1: Cluster-robust SEs from `rdrobust` (61 clusters)
- H2/H3 interactions: Wild cluster restricted (WCR) bootstrap — Cameron, Gelbach & Miller (2008), Webb six-point weights, $B = 99,999$, null imposed
- Additional: exact matching on topic + publication date

Robustness toolkit: placebo tests, leave-one-out, OVB sensitivity [► Details](#)

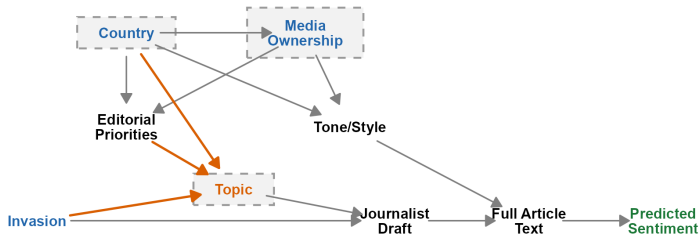
The Identification Problem

A. Controlling for Topic Only
(Backdoor path OPEN)



The Solution: Conditioning on Country, Ownership, Topic

B. Controlling for Country, Ownership, and Topic
(Backdoor path BLOCKED)



D. Results Tables

H1: Robust Across Specifications

	(1) Baseline	(2) Topic FEs	(3) Topic + Source FEs	(4) Z-Score Non-Mil.	(5) No NATO
Invasion	−0.130*** (0.022)	−0.115*** (0.021)	−0.106*** (0.015)	−0.184*** (0.033)	−0.284*** (0.038)

All models use the optimal bandwidth. SEs clustered at source level. Models 4 and 5 use source-level Z-score normalization; Model 4 omits military topics. Model 5 additionally excludes NATO allies.

H2: State Ownership Interaction

	(1) [‡] State RDD	(2) [‡] Indep. RDD	(3) [†] Non-Mil.	(4) [†] Non-NATO	(5) [†] Matched
Invasion	-0.109+ (0.058)	-0.214*** (0.034)	-0.240*** (0.031)	-0.347*** (0.027)	-0.348*** (0.026)
State × Treat			0.14+ (0.06)	0.222* (0.069)	0.222* (0.069)

[‡]Separate RDDs with cluster-robust SEs. [†]WCR bootstrap ($B = 99,999$). (3) omits military topics. (4) omits NATO allies. (5) exact matching on topic + date. All use Z-score normalization and optimal bandwidth.

H2: State Management Varies by Topic

	Political	Military	Economic	Cultural	Other
Invasion	-0.213*** (0.027)	-0.422*** (0.068)	-0.423*** (0.066)	-0.547*** (0.071)	-0.269*** (0.039)
State $\times$ Treat	0.155 (0.079)	0.012 (0.102)	0.290* (0.130)	0.100 (0.190)	0.270** (0.100)

State management strongest in Economic and Other coverage; not significant for Military.
Consistent with strategic topic selection — military topics are harder to spin.

H3: Russia-Aligned Interaction — The Critical Test

	(1) [‡] Aligned RDD	(2) [‡] Neutral RDD	(3) [†] Non-Mil.	(4) [†] Matched
Invasion	−0.220* (0.099)	0.055 (0.101)	0.098 (0.070)	0.098 (0.069)
Russia-aligned × Treat			−0.330* (0.110)	−0.380* (0.110)

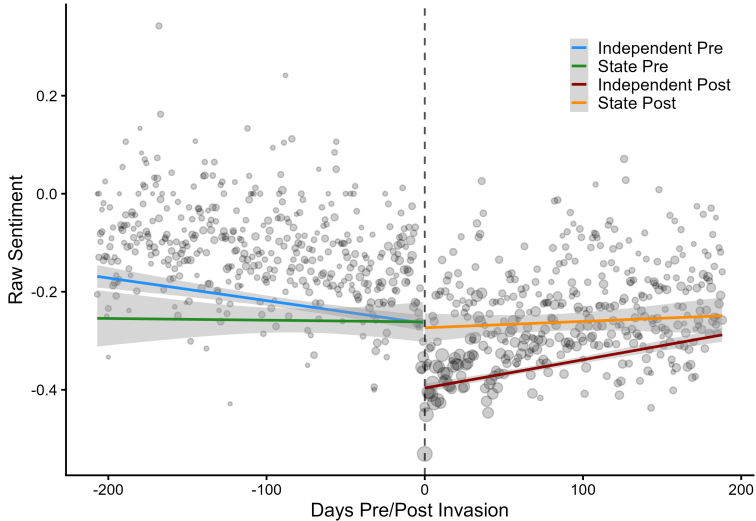
[‡]Separate RDDs with cluster-robust SEs. [†]WCR bootstrap ($B = 99,999$). Sample restricted to state-owned media in non-NATO countries. (3) omits military topics. (4) exact matching on topic + date. All use Z-score normalization and optimal bandwidth.

Ethnic Russian Population: The Null That Matters

	(1) [‡] High RDD	(2) [‡] Low RDD	(3) [†] Optimal BW	(4) [†] Matched
Invasion	-0.079 (0.066)	-0.108 (0.102)	-0.170 (0.100)	-0.143 (0.099)
High Russian × Treat			-0.055 (0.133)	0.015 (0.145)

- If domestic audience management were driving the pattern, ethnic Russian population should matter
- **It doesn't** — interaction is small, unstable in sign, never significant
- Rules out the demographic mechanism; supports the signaling interpretation

Neutral Sentiment: State vs. Independent Media



E. Robustness

Robustness Summary

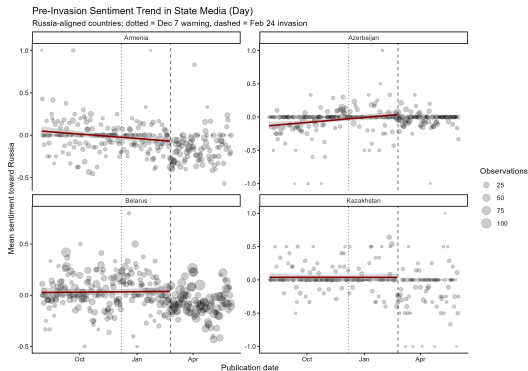
Placebo test: False treatment at Dec 7, 2021 (US public warning) → null

Model	Estimate	SE	p
H1: Full Model (RDD)	0.016	(0.033)	0.637
H1: Z-Score (RDD)	0.000	(0.063)	0.994
H2: Diff-in-Disc	0.250	(0.119)	0.087
H3: Non-Military	0.705	(0.287)	0.143

Additional checks:

- Leave-one-out by country: survives all exclusions (Belarus attenuates) [▶ LOO](#)
- Country-level clustering leaves H2 and H3 significant ($p = 0.021$, $p = 0.032$)
- Exact matching on topic + publication date; OVB sensitivity [▶ OVB plot](#)
- Multiple bandwidth specifications; drop military topics (H3 strengthens)

Pre-Trends and the December Warning



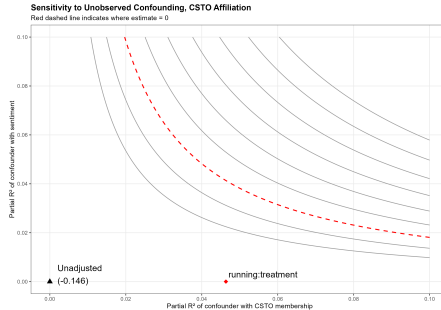
Pre-invasion trends are small and mixed in sign; the only two slopes distinguishable from zero point in opposite directions. At the December 7 placebo cutoff, the pooled estimate is 0.108 ($p = 0.225$); only Belarus is individually significant, and it is *positive* (0.320, $p = 0.025$).

H3: Leave-One-Out Country Robustness

<i>Standard Model</i>				<i>Matched Model</i>			
Dropped	Coeff.	SE	p	Dropped	Coeff.	SE	p
Belarus	−0.188	(0.136)	0.201	Belarus	−0.203	(0.142)	0.186
Armenia	−0.258*	(0.102)	0.030	Armenia	−0.344*	(0.109)	0.010
Georgia	−0.230*	(0.101)	0.043	Georgia	−0.280*	(0.107)	0.024
Azerbaijan	−0.361*	(0.136)	0.024	Azerbaijan	−0.387*	(0.136)	0.017
Kazakhstan	−0.249*	(0.085)	0.015	Kazakhstan	−0.277**	(0.076)	0.004
Moldova	−0.326*	(0.114)	0.015	Moldova	−0.383**	(0.110)	0.005
Kosovo	−0.298	(0.138)	0.054	Kosovo	−0.348*	(0.130)	0.021
Serbia	−0.387**	(0.112)	0.005	Serbia	−0.407**	(0.112)	0.004

Belarus attenuation expected — as the most constrained state, its removal reduces the signal. Result otherwise stable across all exclusions.

OVB Sensitivity Analysis: Russia-Aligned Interaction



Robustness value of 1.9%: an unobserved factor would need to explain at least 1.9% of the residual variance of both group assignment and sentiment to drive the differential to zero. Benchmarked against the invasion indicator at $k_d = 0.25$, a confounder would need a 56% association with the interaction and be 1.5 times as strongly associated with sentiment as the invasion itself.

F. Discussion

What the Findings Mean

States whose formal channels are constrained by dependence on a powerful patron produce systematically different media signals than states whose channels remain open.

- State media sentiment is not noise — it is a legible instrument of international communication
- The signal's value is that it shifts receivers' beliefs about alignment — it doesn't require revealing "true" preferences
- The channel is institutionally buffered below the formal-break threshold, and sustained by mutual dependence between patron and client
- Western trade relationships with Russia-aligned states remained stable through 2022 — consistent with these states not being categorized alongside Russia [▶ Trade Data](#)

Cohen (1987) was right. The channel Soviet-aligned states used during the Cold War remains operative — and is now empirically identifiable.

Western Trade Shares: Russia-Aligned States

Country	2021	2022
Azerbaijan	72%	73%
Kazakhstan	46%	48%
Armenia	34%	29%
Belarus	22%	(disrupted by sanctions)

Western trade shares (Europe + North America as % of total trade) remained broadly stable through 2022. Consistent with the interpretation that these states were not categorized alongside Russia by Western governments — though we cannot establish a direct causal link to the media signal. Source: World Bank (2021, 2022).

Limitations and Extensions

Limitations:

- Single case — but theory generalizes to any patron-client relationship under constraint
- Cannot directly observe signal *receipt* — we identify the sent signal, not the received one
- Factional incoherence vs. strategic coordination: RDD identifies the effect regardless of internal mechanism

Extensions:

- Other patron-client dyads (states in China's security orbit; U.S.–Central America in the 1980s/90s)
- U.S. allies navigating the tension between security dependence and pressure